

Ravisankar Selvaraju

Robotics Application Engineer | Autonomous Systems

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About Me

A robotics enthusiast with a foundation in mechatronics engineering, completing a master's in autonomous systems. I am passionate about robotics from a young age, which led me to pursue both my bachelor's and master's degrees in the field of robotics. I enjoy solving real-time problems as a team under pressure, which motivates me to participate in robotics competitions. Notably, we won the world championship in RoboCup@Work 2023 as the b-it-bots team from H-BRS. I have been working on state estimation and improving odometry in autonomous systems for the past two years. My master's thesis on slip-aware state estimation for a planetary rover using Invariant Extended Kalman Filter was submitted in July 2026.

Education

M.Sc. in Autonomous Systems (GPA: 2.2, thesis submitted July 2026) *September 2021 – July 2026*
Hochschule Bonn-Rhein-Sieg(H-BRS), Germany

B.E. in Mechatronics Engineering (CGPA: 9.13 / 10.0) *August 2017 – April 2021*
Kongu Engineering College, India

Key Skills

State Estimation, Sensor Fusion, Kalman Filtering, Robot Kinematics & Dynamics

C++, Python, ROS, Git, Docker, Linux, SolidWorks 3D modelling (CSWA & CSWP Certified)

Professional Experience

Research Assistant *July 2023 – Present*
Robotic Innovation Center, DFKI Bremen, Germany

- Evaluated and improved odometry algorithms for rimless wheeled rovers in GPS-denied, slip-prone environments using the ROCK framework and MARS simulation.
- Implemented ROS2 AMCL for localization of rimless wheeled rovers in simulated and real-world environments, and assisted in tip-over prediction using deep learning and physics-based models.
- Contributed to the European Rover Challenge (ERC) 2025 qualification, including manipulator simulation, Raspberry Pi driver interface development, and Docker and version control setup for the team.

Teaching Assistant | Developer in b-it-bots @work RoboCup Team *November 2021 – July 2023*
Hochschule Bonn-Rhein-Sieg (H-BRS) Bonn, Germany

- Organized and supported the Foundation Course for Winter Semester 2022, assisting students in mathematics and programming fundamentals.
- Developed and assisted in perception and manipulation modules (empty space detection, pre-grasp planning, task planning) for a ROS Noetic mobile manipulation system for RoboCup@work competitions.
- Contributed to the **bit-bots @work lab** at HBRS using Python and ROS/ROS2 to solve RoboCup competition tasks leading to the world championship victory in 2023 RoboCup@work league in Bordeaux, France.

- Developed **force-controlled manipulation behaviors** (slider task, door opening, probe insertion) with velocity/force feedback for the Robothon 2023 robotics competition as part of the b-it-bots team using Kinova Gen3 arm and secured 6th Place in Grand Challenge 2023.

Engineering Intern - IoT & Data Analytics *November 2020 – March 2021*
Ford India Private Limited India

Built an IoT based CAN data collection and processing system using **Raspberry Pi** for brake data analysis in Ford motor vehicles for driver safety and performance.

Technical Projects

Slip-Aware State Estimation for Rimless Wheeled Rovers using Sensor Fusion (Master's Thesis, Submitted July 2026)

- Fused IMU, Wheel encoder, and contact-based measurements using an Invariant Extended Kalman Filter (InEKF) to estimate rover pose under slip-prone conditions
- Designed the PCV (Potential Contact Velocity) update model with adaptive slip filtering to reject erroneous wheel encoder measurements on loose terrain.
- Validated the approach on the Coyote 3 rover at RIC DFKI using VICON motion capture as ground truth in real-world experiments.

Skills: InEKF, Sensor Fusion, Slip Elimination, State Estimation

Improving Robotic Perception by Kinematics and Dynamics Feed-forward

- Developed a framework to enhance robotic perception by leveraging underutilized kinematics and dynamics solver data from a Kinova Gen3 manipulator, validated experimentally with two use cases.
- Formulated a mapping function from focus value to object distance, enabling real-time Camera Refocusing that outperformed the existing autofocus mechanism during arm motion.
- Implemented ACHD solver-based robot motion control and fused IMU data with solver output via a Kalman filter for Motion State Estimation.

Skills: Robot kinematics, Dynamics solver, C++, Kalman filter, Sensor Fusion

Publications

Prediction-Based Tip-Over Prevention for Planetary Exploration Rovers - *The 14th EASN International Conference, Greece, 2024*

b-it-bots: Winners of RoboCup@Work 2023, Book chapter - *RoboCup 2023: Robot World Cup XXVI, 2024*

Real-Time Performance Analysis of Multiple Parameters of Automotive Sensor CAN Data to Predict Vehicle Driving Efficiency - *International Journal of Computing and Digital Systems, 2022*

IoT-Based Rash Braking Data Analysis and Plotting in Google Maps Using Raspberry Pi - *International Conference on Data Analytics for Business and Industry, 2021*

Notable Achievements

- **RoboCup@Work 2023 - World Champion**, b-it-bots Team, Hochschule Bonn-Rhein-Sieg (H-BRS)
- **6th Place - Robothon Grand Challenge 2023**
- Contributed to **European Rover Challenge (ERC) 2025 Qualification** - Robotics Software System Development